EDS223 - HW2

Exploring patterns of environmental justice

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2026-10-21

Rubric

Data analysis

- Code must produce expected output based on correct data manipulation
- Spatial operations must be performed correctly with appropriate verification (e.g., CRS matching, appropriate join methods, checking results)
- Multiple approaches can be used to reach the correct output. Analysis must demonstrate critical interrogation of the approach used by showing justification and verification of intermediate steps. "Correct" answers are not sufficient. Justification and verification of approach should be demonstrated using the following:
  - custom warning and error message (e.g. warning() and stop(); resources from EDS 221)
  - informative comments (resource from EDS 220)
  - BONUS: unit tests (e.g. using {testthat}; resources from EDS 221)

Plots, tables, and maps

- All plots, tables, and maps must be clear, accurate, and effectively convey the intended information
- All plots and maps must include the following elements:
  - an informative title
  - legends with legible titles, including units
  - color scales that are accessible (i.e. make intuitive sense) and appropriate to the data (i.e. discrete vs. continuous)
  - for maps: indication of scale and orientation (i.e. graticules/gridlines or scale bar and compass), and an appropiate basemap included where necessary
- All tables should be rendered with legible titles and styling (e.g. {gt})

Written reflections

- Reflections must be clear, accurate, and demonstrate a deep understanding of the analysis performed.

Professional Output

- The rendered Quarto doc must show all required elements in a professional style with explanation of each step of analysis
  - see examples of professional and unprofessional output on the Assignments page

Setup

```
# Load packages
library(tidyverse)
```

```
-- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
v dplyr     1.2.1     v readr     2.2.0
v forcats   1.0.1     v stringr   1.6.0
v ggplot2   4.0.3     v tibble    3.3.1
v lubridate 1.9.5     v tidyr     1.3.2
v purrr     1.2.2
-- Conflicts ------------------------------------------ tidyverse_conflicts() --
x dplyr::filter() masks stats::filter()
x dplyr::lag()    masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to beco
```

```
library(here)
```

```
here() starts at /Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW2
```

```
library(sf)
```

```
Linking to GEOS 3.13.0, GDAL 3.8.5, PROJ 9.5.1; sf_use_s2() is TRUE
```

```r
library(tmap)
```

```
Registered S3 method overwritten by 'stars':
  method                   from
  st_interpolate_aw.stars sf
```

```r
library(spData)
library(knitr)
library(ggpubr)

# Set warnings to display as they occur
options(warn = 1)

# Set ggplot theme
set_theme(theme_bw())
```

```r
# Load data
# EJScreen data - environmental and demographic US info at the block group level
ejscreen <- sf::st_read(here("data","ejscreen",
                             "EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb"))
```

```
Reading layer `EJSCREEN_StatePctiles_with_AS_CNMI_GU_VI' from data source
  `/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW2/data/ejscreen/EJSCREEN_2023_BG_StatePct_with_AS_CNMI_GU_VI.gdb'
  using driver `OpenFileGDB'
Simple feature collection with 243021 features and 223 fields
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -19951910 ymin: -1617130 xmax: 16259830 ymax: 11554350
Projected CRS: WGS 84 / Pseudo-Mercator
```

```r
# HOLC Redlining data - HOLC maps and area descriptions for LA
holc <- sf::st_read(here("data","mapping-inequality",
                             "mapping-inequality-los-angeles.json"))
```

```
Reading layer `mapping-inequality-los-angeles' from data source
  `/Users/smsoriano/Documents/Documents - Nawiliwili/UCSB_MEDS/eds223/EDS223-
HW2/data/mapping-inequality/mapping-inequality-los-angeles.json'
  using driver `GeoJSON'
Simple feature collection with 417 features and 14 fields
```

```
Geometry type: MULTIPOLYGON
Dimension:     XY
Bounding box:  xmin: -118.6104 ymin: 33.70563 xmax: -117.7028 ymax: 34.30388
Geodetic CRS:  WGS 84
```

```r
# Bike + Ped Crash data - LA crash records
crashes <- read_csv(here("data","SWITRS_LA","Crashes.csv"))
```

```
Rows: 251100 Columns: 80
-- Column specification --------------------------------------------------------
Delimiter: ","
chr  (45): COLLISION_TIME, OFFICER_ID, REPORTING_DISTRICT, CHP_BEAT_TYPE, CI...
dbl  (33): CASE_ID, ACCIDENT_YEAR, JURIS, DAY_OF_WEEK, CHP_SHIFT, POPULATION...
date  (2): PROC_DATE, COLLISION_DATE

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
# Warning - CRS - confirm crs match, transform if not
if (st_crs(ejscreen)$epsg == st_crs(holc)$epsg){
    # CRS match - true
    print("CRS do match!")
} else {
    # CRS match - false
    warning("CRS do NOT match!")
    print("Changing HOLC data CRS to match EJScreen data CRS EPSG:3857...")
    # CRS transform to match ejscreen data
    ejscreen <- st_transform(ejscreen, crs = st_crs(ejscreen)$epsg)
    holc <- st_transform(holc, crs = st_crs(ejscreen)$epsg)
    print("CRS match complete!")
}
```

```
Warning: CRS do NOT match!
```

```
[1] "Changing HOLC data CRS to match EJScreen data CRS EPSG:3857..."
[1] "CRS match complete!"
```

Part 1: Legacy of redlining in current environmental (in)justice

Question 1: Map of Historical Redlining Neighborhoods

```
# Correct HOLC colors per grade
holc_colors = c("A" = "#76a865",
                "B" = "#7cb5bd",
                "C" = "#f1db11",
                "D" = "#d9838d",
                "NA" = "#000000")
# Create holc polygon map over la county basemap
tm_basemap(server = "Esri.WorldTopoMap") +
tm_shape(holc) +
    tm_polygons(fill = "grade",
                fill_alpha = 0.5,
                lwd = 0.5,
                fill.scale = tm_scale_categorical(values = holc_colors),
                fill.legend = tm_legend(title = "HOLC Grade",
                                        # Specify legend layout
                                        position = tm_pos_out("right"))) +
tm_compass(position = c("left","bottom")) +
tm_scalebar(position = c("left","bottom")) +
tm_options(component.autoscale = FALSE) +
tm_title("Los Angeles County: Redlined Neighborhoods",
            frame = TRUE,
            size = 1.5) +
tm_layout(# Specify legend title size
          legend.title.size = 1.0,
          # Specify legend text size
          legend.text.size = 0.8,
          # Set font family
          text.fontfamily = "sans",
          # Set inner margins to give scalebar room
          inner.margins = c(0.1,0.1,0.05,0.05)
          )
```

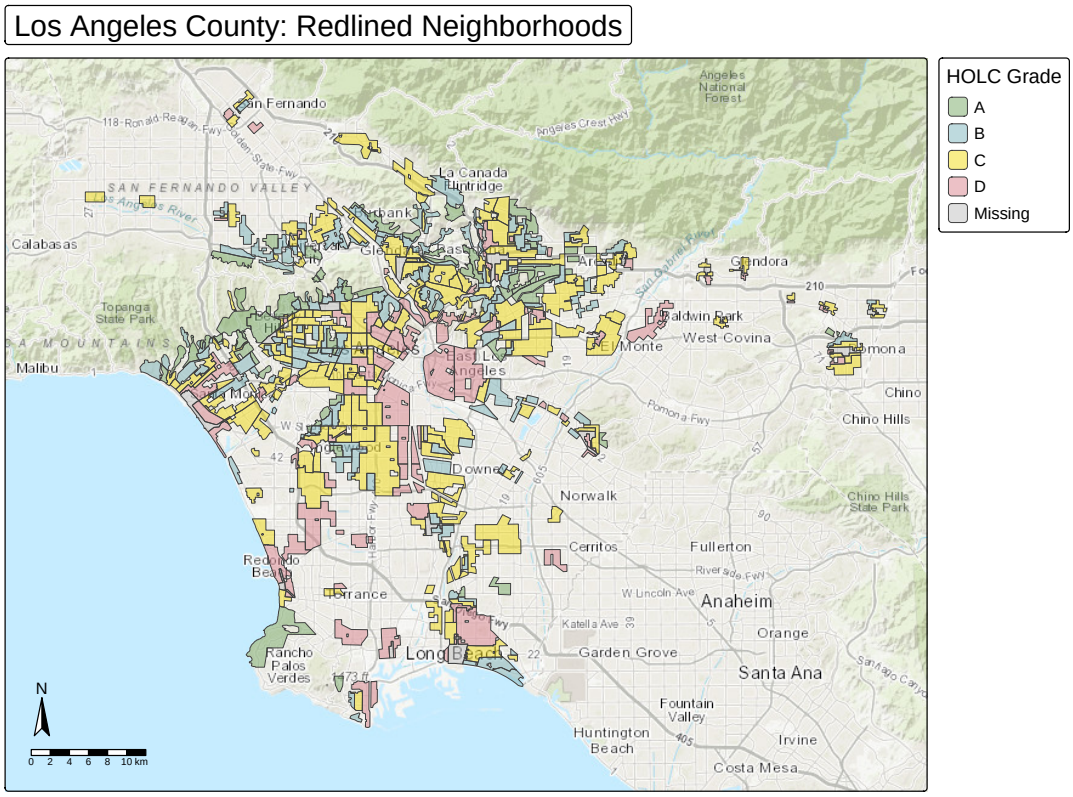


Figure 1: Map of Los Angeles (LA) county in California (CA). Neighborhoods with HOLC grades assigned from 1935-1940 - A, B, C, and D - are represented on this map by color. An "A" HOLC grade is the best "residential security", while a "D" HOLC grade is the worst "residential security."

Question 2: Table of Census Blocks Within HOLC Grades

```
# Filter ejscreen to los angeles county in CA
los_angeles_ejs <- ejscreen |>
    dplyr::filter((ST_ABBREV == "CA") &
                (CNTY_NAME == "Los Angeles County"))

# Confirm st_intersect will work in st_join for ejscreen LA and holc datasets
if(any(st_intersects(los_angeles_ejs, holc, sparse = FALSE))){
  print("st_intersect will work - geometries overlap.")
} else{
```

```
  stop("st_intersect will not work - no overlapping geometries.")
}
```

```
[1] "st_intersect will work - geometries overlap."
```

```
# Join los_angeles_ejs to holc by intersecting geometry
la_holc <- st_join(los_angeles_ejs, holc, join = st_intersects) |>
    # Select columns relevant to Part 1
    select(ID,grade,LOWINCPCT,P_PM25,P_LIFEEXPPCT) |>
    # Drop geometries
    st_drop_geometry()

# Create summary table of census block (cb) counts and percentages per holc grade
la_holc_graden <- la_holc |>
    group_by(grade) |>
    summarise(cb_count = n()) |>
    mutate(cb_perc = (cb_count / nrow(la_holc)) * 100)

# Use kable to format table output
kable(la_holc_graden,
        col.names = c("HOLC Grade","Census Blocks per Grade (Count)",
                      "Census Blocks per Grade (Percent)"))
```

Table 1: Counts and percentages of EJScreen census blocks per HOLC Grade in LA county. The percentages were calculated using a total LA county census block total of 8988 (see Results Summary for details).

HOLC Grade	Census Blocks per Grade (Count)	Census Blocks per Grade (Percent)
A	449	4.99555
B	1239	13.78505
C	3058	34.02314
D	1346	14.97552
NA	2896	32.22074

Question 3: EJScreen Data Visualizations Within HOLC Grades

```r
# Create df for visualizations
la_holc_sum <- la_holc |>
    # Group by grade
    group_by(grade) |>
    # Calculate means by grade
    summarise(mean_LOWINCPCT = mean(LOWINCPCT, na.rm = TRUE),
              mean_P_PM25 = mean(P_PM25, na.rm = TRUE),
              mean_P_LIFEEXPPCT = mean(P_LIFEEXPPCT, na.rm = TRUE)) |>
    rename("HOLC Grade" = "grade")
```

```r
# Generate plots

# Generate bar plot - mean_LOWINCPCT across HOLC grades
sum_plot1 <- ggplot(la_holc_sum,
    aes(x = `HOLC Grade`,
        y = mean_LOWINCPCT*100,
        fill = `HOLC Grade`)) +
    geom_bar(stat = "identity") +
    scale_fill_manual(values = holc_colors) +
    theme(axis.title.x = element_blank(),
          axis.text.x  = element_blank()) +
    labs(title = "Low Income",
         y = "Mean %")

# Generate bar plot - mean_P_PM25 across HOLC grades
sum_plot2 <- ggplot(la_holc_sum,
    aes(x = `HOLC Grade`,
        y = mean_P_PM25,
        fill = `HOLC Grade`)) +
    geom_bar(stat = "identity") +
    scale_fill_manual(values = holc_colors) +
    theme(axis.title.x = element_blank(),
          axis.text.x  = element_blank()) +
    labs(title = "Particulate Matter 2.5",
         y = "Mean US Percentile")

# Generate bar plot - mean_P_LIFEEXPPCT across HOLC grades
sum_plot3 <- ggplot(la_holc_sum,
    aes(x = `HOLC Grade`,
        y = mean_P_LIFEEXPPCT,
        fill = `HOLC Grade`)) +
    geom_bar(stat = "identity") +
```

```
    scale_fill_manual(values = holc_colors) +
    theme(axis.title.x = element_blank(),
          axis.text.x  = element_blank()) +
    labs(title = "Life Expectancy",
         y = "Mean US Percentile")

# Arrange bar plots in one figure
ggarrange(sum_plot1, sum_plot2, sum_plot3, ncol = 3, nrow = 1,
            labels = c("(a)", "(b)", "(c)"),
            font.label = list(face = "plain"),
            common.legend = TRUE,
            legend = "bottom") |>
# Add figure title
annotate_figure(top = text_grob("HOLC Neighborhoods: Current EJScreen Statistics",
                                size = 20))
```

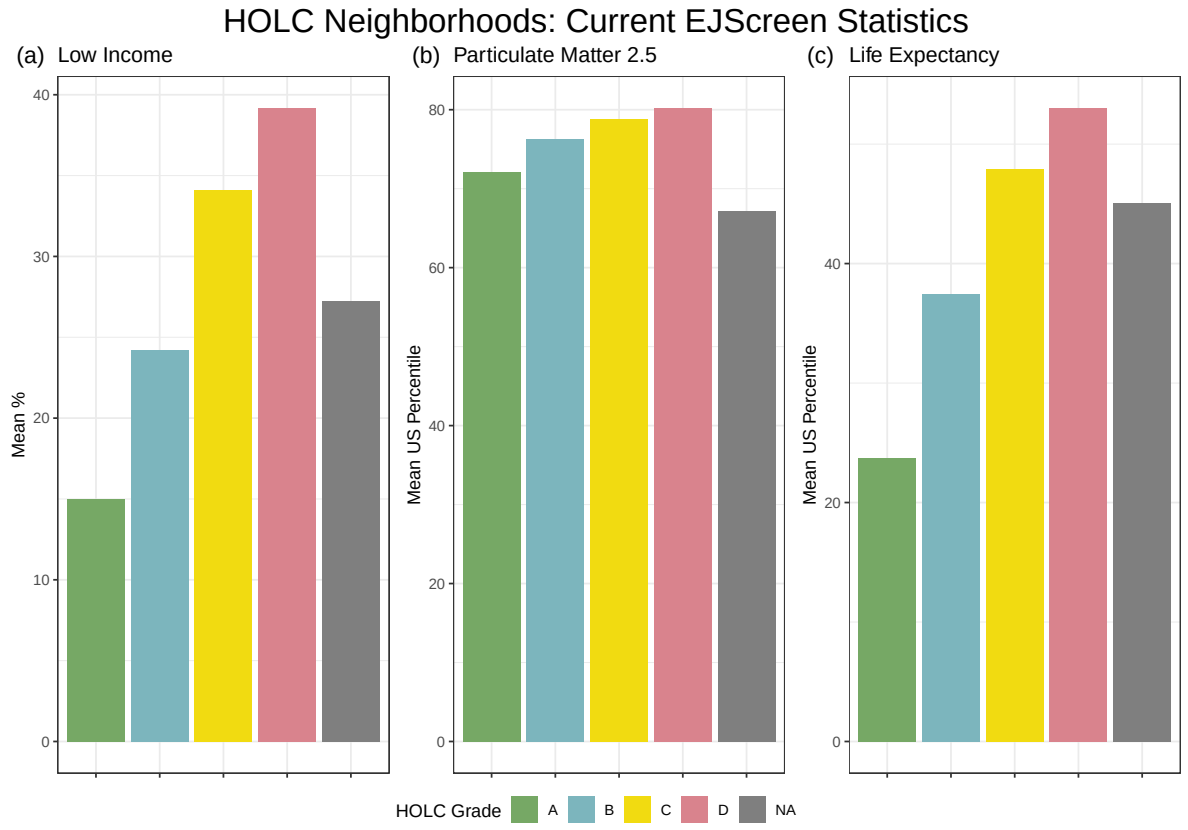


Figure 2: Bar plots showing current (2023) EJScreen data for three statistics - (a) low income (%), (b) US percentile for Particulate Matter (PM) 2.5, and (c) US percentile for low life expectancy - each averaged across historical HOLC neighborhood (redlining) grades.

Question 4: Results Summary

For LA county, the number of EJScreen census blocks within a Home Owners' Loan Corporation (HOLC) grade is not evenly distributed across the four grades - "A", "B", "C", "D". The "A" and "B" grades combined only have around 19% of the total LA county census blocks noted in EJScreen, while grade "C" has about 34% and grade "D" has 15% (32% of the census blocks were not associated with a grade) (*Table 2*). It should be noted that the total number of EJScreen census blocks in LA county was 6591, this number was increased to 8988 due to 1731 cases of multiple HOLC grades appearing in a single census block. To preserve HOLC data, data from a single census block may be used in calculating summary statistics for more than one HOLC grade.

When comparing the HOLC grade means of three metrics recorded per census block in EJScreen - low income (%), US percentile for Particulate Matter (PM) 2.5, and US percentile for low life expectancy - a similar trend was observed across all three comparisons. The means calculated for low income (%), percentile for PM2.5, and percentile for low life expectancy all increase sequentially from grade "A" to grade "D" (*Figure 2*). This implies that historical redlining by HOLC from 1935-1940 is possibly correlated with current (2023 EJScreen) trends in low income, PM2.5, and low life expectancy. In other words, the census blocks located in grade "A" HOLC neighborhoods currently have the lowest mean percentage of low income households (14.89%), and fall in lower mean percentiles than the other three grades for annual PM2.5 air exposure (~72th percentile) and low life expectancy (~24th percentile). In contrast, grade "D" HOLC neighborhoods have the highest mean percentage of low income households (39.19%), and fall in higher mean percentiles than the other three grades for annual PM2.5 air exposure (~80th percentile) and low life expectancy (~53rd percentile). An alternative interpretation of the US percentile metric is that if a group - like grade "D" - is in the 53rd US percentile for low life expectancy, then only 47% of the US has a higher low life expectancy metric. Given these results, it appears that historically redlined "C" and "D" grade census blocks do currently experience more negative environmental and socioeconomic conditions than census blocks in historical HOLC "A" and "B" grade neighborhoods.

Part 2: Legacy of redlining in pedestrian and bike accidents

Your second task is to explore the legacy of historical redlining in Los Angeles on the report of pedestrian and bike crashes/ fatalities.

For this assignment, you must produce the following based on observations from 2020-2025:

1. A figure summarizing the percent of observations within redlined neighborhoods within each HOLC grade
    - Create a visualizations that shows:
    - The percentage of bike and pedestrian crashes and fatalities within each HOLC grade

- Include an appropiate title, axis labels, and legend
- Hints: Make sure to filter for bike and pedestrian crahes(PEDESTRIAN_ACCIDENT and BICYCLE_ACCIDENT). Look at the difference in NAs between POINT_X/POINT_Y and LONGITUDE/LATITUDE before you decide which to use. Ensure the SWITRS data and HOLC dataset have matching CRS', then perform a spatial join to assign each crash to a corresponding HOLC grade.

Data

The data used for this assignment is from the EJScreen: Environmental Justice Screening and Mapping Tool, which no longer exists online. However, our data can be accessed through the following link: https://drive.google.com/file/d/1nG6Nj1bXfzQFOVMO8Km3eNy4SWu1YcIQ/view

References

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